DonProof PU-C100

Cold applied pure polyurea membrane for waterproofing and protection

DESCRIPTION

DonProof PU-C100 is a two-component, high solids, fast curing, aromatic pure polyurea coating that cures to form an elastomeric watertight and waterproof polyurea membrane, with high mechanical properties and strong adhesion to many types of surfaces.

DonProof PU-C100 adaptability and drying time makes it suitable for application on uneven surfaces and in areas of any shape, whether curved or squared.

APPLICATIONS

Waterproofing and protection of:

- Wet areas, bathrooms and kitchens.
- Balconies and roof decks.
- Under-tile applications,
- Concrete structures.
- Floors, parking lots and garages.
- Over asphalt membranes.
- Over polyurethane insulation foams
- Light roofing made of metal or fibrous cement.
- Green roofs and planters.

ADVANTAGES

- Excellent mechanical properties, high tensile and tear strength, high abrasion resistance.
- Excellent adhesion to most common substrates.
- Strong, flexible and hard-wearing membrane.
- Cold applied with a roller, brush, trowel, squeegee or spray.
- Excellent stability and durability.
- Fast curing.
- Friendly to the environment, low VOC compounds.
- Good breathability characteristics which minimize the accumulation of humidity under the coat.
- Optimum maintenance and cleaning properties for the cured surface.
- Resistant to root penetration.

TECHNICAL PROPERTIES @ 25°C:	
Colour:	Grey, white, as requested
Density:	1.30 ± 0.05 g/cm³
Solid content:	91 ± 2%
Pot life:	70 min@ 25°C 50 min @ 40°C
Tack free time:	3:30 - 4:30 hr
Minimum recoating time:	4 hr
Maximum recoating time:	7 days
Full cure:	7 days
Application temperature:	5 to 35°C
Service temperature:	-30 to 80°C
Shore hardness: ASTM D2240 Shore A	60 ± 5 @ 7 days
Taber abrasion: ASTM D4060 (1000 g, 1000 cycle) CS17 wheel	≤ 50 milligram
Tensile strength: ASTM D412	≥ 3.5 MPa
Elongation: ASTM D412	≥ 450%
Bond strength to concrete: ASTM D4541	≥ 1.5 MPa

DonProof PU-C100

METHOD OF USE

SURFACE PREPARATION

The surface should be clean, dry, sound, and free from oil, grease, and wax contamination. Cement laitance, loose particles, mold release agents, or curing membranes must be removed.

The surface must have a minimum compression strength of 25 MPa and a minimum adhesion strength of 1.5 MPa.

Fill surfaces irregularities with a suitable epoxy resin mortar. The maximum atmospheric relative humidity should not exceed 85%. New concrete structures need to dry for at least 28 days. Do not proceed with the application in the presence of moisture.

PRIMING

Concrete substrates:
It is recommended to prime absorbent surfaces such as concrete, cement or screed substrates using Donproof Primer PU.

For damp surfaces where the relative humidity level of the screed is higher than 75%, Donproof DPM should be applied by brush or short-hair lambswool roller.

Non-porous substrates:
Steel, Aluminum, ceramic and porcelain tiles, should be primed using Donproof Primer PU.

Consult DCP Technical Department for specific primer recommendation based on the project conditions.

Avoid any primer ponding on the floor. Highly porous surfaces may require two coats of primer.

Refer to the Technical Datasheet of the primer for further information.

MIXING

Prior to mixing, stir the individual components of DonProof PU-C100 for 2 - 3 minutes using a low speed mixer.

Taking care to ensure that the bottom and sides are thoroughly scraped. Transfer the entire contents of component B into component A container and mix slowly for at least 3 minutes until a homogeneous mix is achieved.

TECHNICAL PROPERTIES @ 25°C:	
Tear strength: ASTM D624	≥ 20 N/mm
Crack Bridging: ASTM C1305 @ 23°C & 20°C	no cracks occurred after 10 cycles @ 1.5 mm
Resistance to water vapor diffusion: EN 1931	μ=1.700
Resistance to roots: PD CEN/TS 14416:2014	No penetration
Water vapor transmission: ASTM E96	Medium 0.6 to 6.0 g/(m²/hr)
Water permeability: Din 1048 @ 5 bars	Nil (Pass)
VOC:	≤ 130 g/ltr

APPLICATION

Use brush, squeegee, lambs wool roller, or airless spray machine to apply the mixed DonProof PU-C100 onto the prepared and primed surface.

If needed, a second coat should be applied at a right angle to the first coat after 4 hours to 7 days.

CONSUMPTION

DonProof PU-C100 should be applied in one or two coats with minimum total consumption of 1.8 - 2.0 kg/m² to provide 1.5 mm total dry film thickness.

UV PROTECTION COAT

As with all aromatic coatings, DonProof PU-C100 should be sealed with a suitable polyurethane protection coat to prevent yellowing due to UV rays exposure. However, this will not cause any negative effect on the physical properties of the product

Consult DCP Technical Department for specific recommendations for an aliphatic topcoat to guarantee its durability.

DonProof PU-C100

REMARKS

- When exposed to UV, a slight discoloration of the membrane is probable. This will not affect, in any way, the waterproofing properties of the final coat.
- DonProof PU-C100 should not be applied at temperatures below 5°C. The surface to be treated and the uncured product must be 3°C above the dew point.
- The curing process occurs more quickly at higher temperatures and slower at lower temperatures.
- At low temperatures extra care is required when storing and mixing DonProof PU-C100, before mixing and processing ensure that the product is stored at temperature no less than 15°C, low temperatures result in high viscosity (thicker liquid) and will lead to poor mixing and failure in application delivery.

CLEANING

Clean all tools after finishing with DCP Solvent PU.

PACKAGING

DonProof PU-C100 is available in 23 kg packs (component A (21.15 kg) and component B (1.85 kg).

STORAGE

DonProof PU-C100 has a shelf life of 12 months from date of manufacture if stored in a dry place, away from direct sunlight, extreme heat, cold, or moisture and in its original unopened pails at temperatures between 5°C and 30°C.

If these conditions are exceeded, contact DCP Technical Department for advice.

CAUTIONS

HEALTH AND SAFETY

Apply in well ventilated areas. Do not smoke. Do not apply near naked flames. In closed areas use force ventilation and carbon active masks.

For further information, refer to the Material Safety Data Sheet.

DonProof PU-C100

MORE FROM DON CONSTRUCTION PRODUCTS

A wide range of construction chemical products are manufactured by DCP which include:

- Concrete admixtures.
- Surface treatments
- Grouts and anchors.
- Concrete repair.
- Flooring systems.
- Protective coatings.
- Sealants.
- Waterproofing.
- Adhesives.
- Tile adhesives and grouts.
- Building products.
- Structural strengthening.

Note:
We endeavour to ensure that any information, advice or recommendation we may give in product literature is accurate and correct. However, because we have no control over where and how products are applied, we cannot accept any liability arising from the use of the products.